Arithmetic and combinatorics of the pure-X coefficients
An exact scan through genus ten

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The scan covers 144 pure-X coefficients and 16,384 nonzero monomial coefficients of the physical quintic polynomials $P_2, \ldots, P_{10}$. The count 16,384 is simply the observed total; no significance is assigned to its being a power of two. Genera nine and ten were computed after the first patterns had been identified in genera two through eight. All arithmetic and comparisons below are exact. The leading-edge formulas are derived consequences of the imposed BCOV and conifold equations. The broader denominator bound remains a conjectural interpretation of the finite data.

1 What is being scanned

Retain the original YY conventions of the implementation:

$$P_g = C^{g-1}F_g = P_g^{\mathrm{an}} + a_g(X), \qquad a_g(X) = \sum_{n=0}^{d_g} a_{g,n}X^n, \qquad d_g = 3g-3,$$

where $P_g^{\mathrm{an}}(v_1 = v_2 = v_3 = 0, X) = 0$. The coefficient of X^n in the full polynomial is generally a polynomial in v_1, v_2, v_3. The phrase "pure-X coefficient" here means specifically $[u^0 v_1^0 v_2^0 v_3^0 X^n]P_g$. For denominator tests, the scan also examines all the mixed monomials.

Set $x = z/\mu = \psi^{-5}$, $\delta = 1 - x$, and $X = 1 - \delta^{-1}$. The rational coefficients in this normalized ring do not depend on μ. A coordinate conversion back to z introduces known external powers of μ. Consequently keeping μ free removes coordinate-scale clutter, but does not itself create or alter the arithmetic patterns below. All statements are specific to the stated generator and amplitude normalization.

2 The highest pure-X coefficient: an exact formula

Define $b_g = |B_{2g}|$. The result is

$$\boxed{A_g := a_{g,3g-3} = 5^{3g-3}\frac{b_g}{2g(2g-2)}.} \tag{1}$$

The normalized values are in Table 1. The apparently exceptional numerators 691 at genus six and 3617 at genus eight are precisely the Bernoulli numerators, not new unexplained integers introduced by the solver.

Here is the derivation, including the signs. In the rational conifold frame,

$$f_c = 1 + O(\delta^3), \quad \tau = \nu/f_c = \delta + \tfrac{7}{10}\delta^2 + O(\delta^3), \quad K_g = \left[-\frac{\delta f_c^2}{625}\right]^{g-1}.$$

The physical amplitude is $\mathcal{F}_g^c = K_g P_g^{\mathrm{hol}}$, and the gap is

$$\mathcal{F}_g^c = \kappa_g \tau^{2-2g} + O(1), \qquad \kappa_g = \frac{5^{1-g}b_g}{2g(2g-2)}.$$

g	$3g-3$	$a_{g,3g-3}/5^{3g-3}$
2	3	$\frac{1}{240}$
3	6	$\frac{1}{1008}$
4	9	$\frac{1}{1440}$
5	12	$\frac{1}{1056}$
6	15	$\frac{691}{327600}$
7	18	$\frac{1}{144}$
8	21	$\frac{3617}{114240}$
9	24	$\frac{43867}{229824}$
10	27	$\frac{174611}{118800}$

Table 1: Highest pure-X coefficient with the known power of five removed. Every row agrees exactly with the independently computed physical polynomial.

The generators v_i are regular in this frame. Weighted degree implies that only $A_g X^{d_g}$ contributes to the most singular pole. Since $X^{d_g} = (-1)^{d_g}\delta^{-d_g}(1-\delta)^{d_g}$, its sign after multiplication by K_g is $(-1)^{d_g+g-1} = (-1)^{4g-4} = +1$. Thus $A_g/625^{g-1} = \kappa_g$, which gives (1). This is a consequence of the gap normalization already imposed, rather than an independent new boundary condition.

2.1 Determination from lower-order data

Even the Bernoulli input has an exact convolution in lower indices:

$$\boxed{b_1 = \frac{1}{6}, \qquad b_g = \frac{1}{2g+1}\sum_{h=1}^{g-1}\binom{2g}{2h} b_h b_{g-h}, \quad g \geq 2.} \tag{2}$$

This is the classical Bernoulli convolution with its alternating signs removed. One way to derive it is to use

$$H(t) = \frac{t}{2}\cot\frac{t}{2} = 1 - \sum_{g\geq 1}\frac{b_g}{(2g)!}t^{2g}, \qquad tH' = H - H^2 - \frac{t^2}{4}.$$

At degree $2g \geq 4$, equating coefficients gives $(2g+1)b_g/(2g)! = \sum_{h=1}^{g-1} b_h b_{g-h}/((2h)!(2g-2h)!)$. The scan checks this convolution against a separate linear Bernoulli recurrence through $g = 53$. Formula (1) therefore predicts this edge of the polynomial through genus 53 without computing full genus-53 amplitudes.

3 A second exact law: the next pure-X coefficient

Put

$$k_h = 5^{3h-2}\frac{b_h}{2h}, \quad h \geq 1; \qquad k_1 = \frac{5}{12}.$$

Then

$$\boxed{a_{g,3g-4} = -\frac{8}{5}(g-1)A_g + 5(2g-3)k_{g-1} + \sum_{h=1}^{g-1} k_h k_{g-h}.} \tag{3}$$

This is an actual genus-splitting convolution. It matches every genus $2, \ldots, 10$, including the fresh genus-nine and genus-ten computations. For example it gives $a_{2,2} = 205/144$, $a_{3,5} = 33125/1008$,

and $a_{4,8} = 18453125/12096$. It also predicts the sign change in $a_{g,3g-4}$ between genera four and five.

To derive it, let $P_h^{(1)}$ be the normalized one-point vertex and define

$$M_g = [v_1 X^{3g-4}] P_g.$$

At $u = v_1 = v_2 = v_3 = 0$, the highest pure-X term of $P_h^{(1)}$ is $k_h X^{3h-2}$. For $h \geq 2$, differentiating $A_h X^{3h-3}$ gives $5(3h-3)A_h X^{3h-2}$; the connection contributes $-5(h-1)A_h X^{3h-2}$. Their sum is $10(h-1)A_h = k_h$. The same formula includes the separately known $h = 1$ seed $5X/12$. A second covariant derivative has leading pure-X coefficient $5(2h-1)k_h$. Therefore the u^0 anomaly equation gives

$$M_g = -\frac{1}{2}\left[5(2g-3)k_{g-1} + \sum_{h=1}^{g-1} k_h k_{g-h}\right]. \tag{4}$$

This mixed coefficient is also checked directly against every full polynomial.

At the conifold $v_1(0) = 2$. Weighted degree shows that the next pole of P_g can receive contributions only from $A_g X^{d_g}$, $a_{g,d_g-1} X^{d_g-1}$, and $M_g v_1 X^{d_g-1}$. The next term in $K_g P_g$ is consequently

$$625^{1-g}\big[-d_g A_g - a_{g,d_g-1} - 2M_g\big]\delta^{3-2g}.$$

Meanwhile $\tau^{2-2g} = \delta^{2-2g}[1 - \frac{7}{5}(g-1)\delta + O(\delta^2)]$. Equating these coefficients gives $a_{g,d_g-1} = -\frac{8}{5}(g-1)A_g - 2M_g$, hence (3). This derivation uses only the leading coefficients of earlier genera and the specified local frame, without fitting a sequence to the data.

4 A triangular binomial rule determines a larger block

There is a useful extension beyond these two explicit formulas. Given the known anomaly body at genus g, form

$$R_g(\delta) = \frac{\kappa_g \tau^{2-2g}}{K_g} - P_g^{\mathrm{an}}(v_1^c, v_2^c, v_3^c, X_c), \qquad r_{g,j} = [\delta^{-j}] R_g(\delta).$$

The regular $O(1)$ part of the physical amplitude, divided by K_g, is $O(\delta^{1-g})$. Thus $r_{g,j}$ is prescribed for every $j \geq g$. But $X = 1 - \delta^{-1}$ implies

$$[\delta^{-j}] a_g(1-\delta^{-1}) = (-1)^j \sum_{n=j}^{3g-3} \binom{n}{j} a_{g,n}.$$

Consequently there is an exact downward recursion

$$\boxed{a_{g,j} = (-1)^j r_{g,j} - \sum_{n=j+1}^{3g-3} \binom{n}{j} a_{g,n}, \quad j = 3g-3, \ldots, g.} \tag{5}$$

It determines the highest $2g-2$ pure-X coefficients. Unlike the special formula for the next coefficient, the general rule uses the full lower-genus amplitudes through the anomaly body, not just their pure-X subsequences. An independent implementation of this triangular rule predicts and checks 90 coefficients across genera two through ten, without using those unknown coefficients in constructing the predictions. The remaining low powers still require the other boundary data.

5 The constant term is also controlled

The MUM constant-map condition fixes

$$\boxed{a_{g,0} = 100\,625^{g-1}\frac{|B_{2g}B_{2g-2}|}{2g(2g-2)(2g-2)!}.} \tag{6}$$

It is positive in this normalization. This supplies another exactly predictable coefficient, with a product of Bernoulli numbers and a factorial. It is useful to keep this known contribution visible when searching for arithmetic novelty in denominator tables.

6 A proved sieve and an empirical one

6.1 Bernoulli denominators: an established theorem

The von Staudt–Clausen theorem gives

$$\operatorname{den}(B_{2g}) = \prod_{\substack{p \text{ prime} \\ p-1\mid 2g}} p.$$

Together with (1), this completely explains the denominator of the leading coefficient. If $B_{2g} = N_g/D_g$ is reduced, then

$$\operatorname{den}(A_g) = \frac{2g(2g-2)D_g}{\gcd\big(5^{3g-3}|N_g|,\, 2g(2g-2)D_g\big)}.$$

The exact numerator still contains Bernoulli arithmetic, but it is generated by (2). This is a precise sense in which both combinatorics and a prime-divisibility sieve control the same coefficients. When studying denominators, dividing out 5^{3g-3} is transparent. Dividing by the entire leading coefficient would insert its Bernoulli numerator into other denominators and can create misleading appearances of primes such as 691.

6.2 Full coefficient arrays: observed Todd-type bound

Let L_g be the least common multiple of all reduced coefficient denominators of P_g. Let $L_g^{\geq 7}$ retain only its prime factors $p \geq 7$. For every computed genus $2 \leq g \leq 10$, we find the exact equality

$$L_g^{\geq 7} = \prod_{\substack{p\geq 7 \\ p \text{ prime}}} p^{\lfloor 2g/(p-1)\rfloor}. \tag{7}$$

The pure-X coefficient LCM has the same $p \geq 7$ part. Every coefficient is also 5-integral: its denominator is coprime to 5. No denominator prime larger than $2g+1$ occurs in the data.

The product in (7) is the $p \geq 7$ part of the Todd/Hirzebruch denominator of degree $2g$. The Hirzebruch denominator formula is a theorem for Todd polynomials [2]; its applicability as a bound to all our BCOV coefficients has not been proved here. The natural conjecture suggested by the data is

$$\operatorname{ord}_p(L_g) \leq \left\lfloor \frac{2g}{p-1} \right\rfloor, \qquad p \geq 7,$$

with equality throughout the observed range. We do not claim eventual equality for every genus, nor a proof of this upper bound.

There are two qualifications to the apparent strength of this observation. First, primes 2 and 3 have larger exponents than the naive Todd degree-$2g$ formula in some genera, while powers of 5 are absent. The generator normalization and genus-one seed matter. Second, throughout

g	Terms	$\mathrm{ord}_2(L_g)$	$\mathrm{ord}_3(L_g)$	$L_g^{\geq 7}$
2	14	5	2	1
3	62	7	4	7
4	185	11	5	7
5	435	13	6	$7 \cdot 11$
6	877	16	8	$7^2 \cdot 11 \cdot 13$
7	1590	18	9	$7^2 \cdot 11 \cdot 13$
8	2666	23	10	$7^2 \cdot 11 \cdot 13 \cdot 17$
9	4211	25	13	$7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 19$
10	6344	28	14	$7^3 \cdot 11^2 \cdot 13 \cdot 17 \cdot 19$

Table 2: Exact denominators in the full YY polynomials. The factors at 2 and 3 are displayed separately; the last column is the observed $p \geq 7$ part.

$2 \leq g \leq 10$, the single known constant-map coefficient $a_{g,0}$ already attains all the maximal exponents at primes $p \geq 7$. Thus the new empirical information in the rest of the array is the upper bound: none of the other 16,000-plus coefficients introduces a larger exponent. The observed LCM equality is not an independent determination of their numerators.

The bound would still be useful. Polynomial primitives can divide by exponents as large as $3g-3$; the absence of primes between $2g+1$ and $3g-3$ therefore requires cancellations. For example, at genus ten the degree bound permits 27, but no denominator factor 23 appears. A proof would have to explain those cancellations in the normalized recursion, perhaps through a stable-graph or characteristic-class organization. This is a research direction suggested by the data, not an established identification.

A weaker additional observation is

$$\min_{\text{nonzero coefficients } c \text{ of } P_g} \mathrm{ord}_5(c) = \begin{cases} 1, & 2 \leq g \leq 6, \\ 2, & 7 \leq g \leq 10. \end{cases}$$

The staircase $\lceil (g-1)/5 \rceil$ fits these values, but only one jump has been seen. It is recorded as a low-confidence extrapolation, not promoted to a law. It predicts the next increase at genus twelve.

7 Searches that did not give a predictive law

The scan also tested first-order rational-ratio recurrences

$$a_{j+1}/a_j = P(j)/Q(j), \qquad \deg P, \deg Q \leq 2.$$

They were tested both across genus along the top four diagonals (after removing 5^{3g-3}) and across X powers within each available row. Fits used the first five consecutive ratios. Every fitted model failed at held-out coefficients. In particular, a degree-$(2,2)$ rational function can interpolate five ratios, so fitting those five data points alone is not evidence. This is only a bounded search; it does not exclude other recurrence structures.

We have not found a compact new rule fixing all interior numerators from nearby pure-X coefficients. They remain controlled by the full anomaly and boundary system. The evidence supports separating the known Bernoulli and binomial structures from an integer numerator array before attempting further recurrence or congruence searches.

8 Reproduction and precise scope

Run from the extracted directory:

```
python scan_coefficients.py physical/amplitudes.json --output results
python verify_triangular.py
```

The first script uses only the Python standard library. The second uses the bundled snapshot of the exact algebra and period modules, also without external runtime dependencies. Input data, per-genus physical diagnostics for the newly computed genera, and the complete exact results are included. The file `exact_edge_predictions_through_53.json` contains formulas (1) and (3), the constant term, and the mixed coefficient M_g through genus 53. It does not contain full amplitudes above genus ten. The scan checks 45 edge/arithmetic identities against the nine physical polynomials, the Bernoulli convolution through index $g = 53$, and 90 independent triangular coefficient predictions.

References

[1] S. Yamaguchi and S.-T. Yau, *Topological String Partition Functions as Polynomials*, hep-th/0406078, especially the polynomial anomaly equations and the stable-graph discussion.

[2] V. M. Buchstaber and A. P. Veselov, *Todd polynomials and Hirzebruch numbers*, arXiv:2310.07383v4. This paper discusses both the Hirzebruch denominator formula and the von Staudt–Clausen formula, together with combinatorial representations.